AD152A Datasheet

Zhuhai Jieli Technology Co.,LTD

Version：V1.3

Date：2024.11.20

ZHUHAI JIELI TECHNOLOGY CO.,LTD

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AD152A Features

CPU Core

- 32-bit CPU,Built-in ICACH, can be connected to Flash for expansion of code
- The main frequency is up to 120MHz

Memory

- On-chip SRAM 20kbyte
- Built-in Nor Flash memory

Clock Source

- RC Clock frequency about 16MHz
- LRC(low power RC) clock frequency about 200KHz

Digital I/O

- Up to 13 programmable digital I/O pins
- General the IO supports pull-up(10k),pull-down(60k), strong,weak output,input and high impedance
- Up to 12 external interrupt/wake-up source(low power available,can be multiplexed to any I/O, with hardware filter)
- Input channel and Output channel, provide arbitrary IO input and output options for some modules

Digital peripherals

- Two UART Controllers(UART0/1) supports DMA and Flow Control
- Two SPI Controllers with DMA(SPI0/1) support master mode and slave mode
- Built-in Spi Flash to run code
- One SD host controller
- Three 32-bit Asynchronous Divider Timers
- One IIC Controller
- Four channel PWM output
- Infrared remote control decoder
- Watchdog

Analog Peripherals

- 0.5 watt Class-D audio amplifier output
- 10-bit high precision ADC
- Low voltage protection
- Power on reset

PMU

- Soft off current: <2uA
- Power down current: <30uA
- Music mode: <6mA
- LVD range(3bit):1.8V~2.5V, step100mV
- VBAT range : 2.0V~5.5V
- VDDIO range : 2.0V~3.4V

Temperature

- Operating temperature: -40℃ to +85℃
- Storage temperature: -65℃ to +150℃

Package

- SOP16

Application

- Sound Toy
- Audio player

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1 Block Diagram

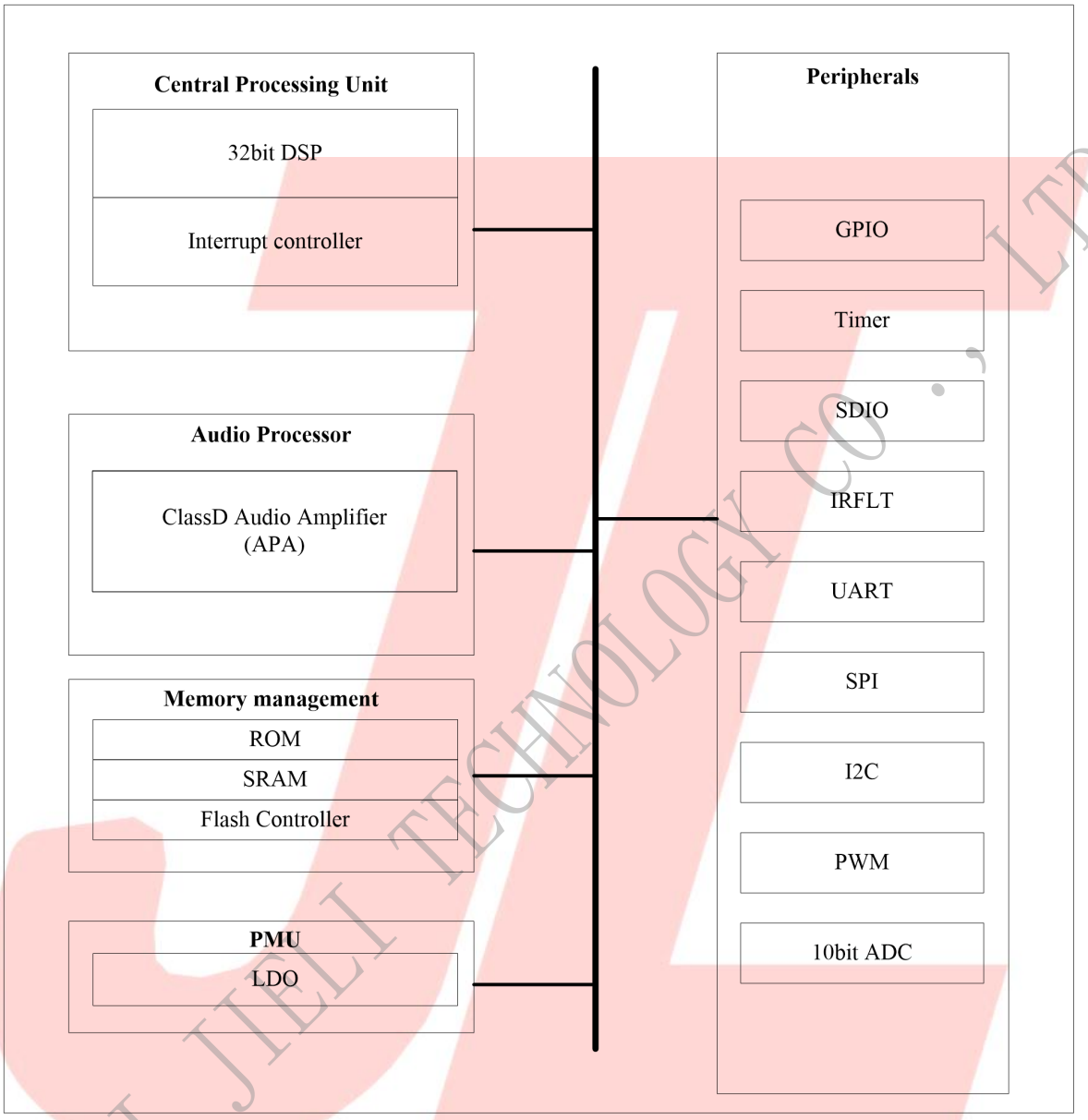

Figure 1-1 AD152A Block Diagram

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2 Pin Definition

2.1 Pin Assignment

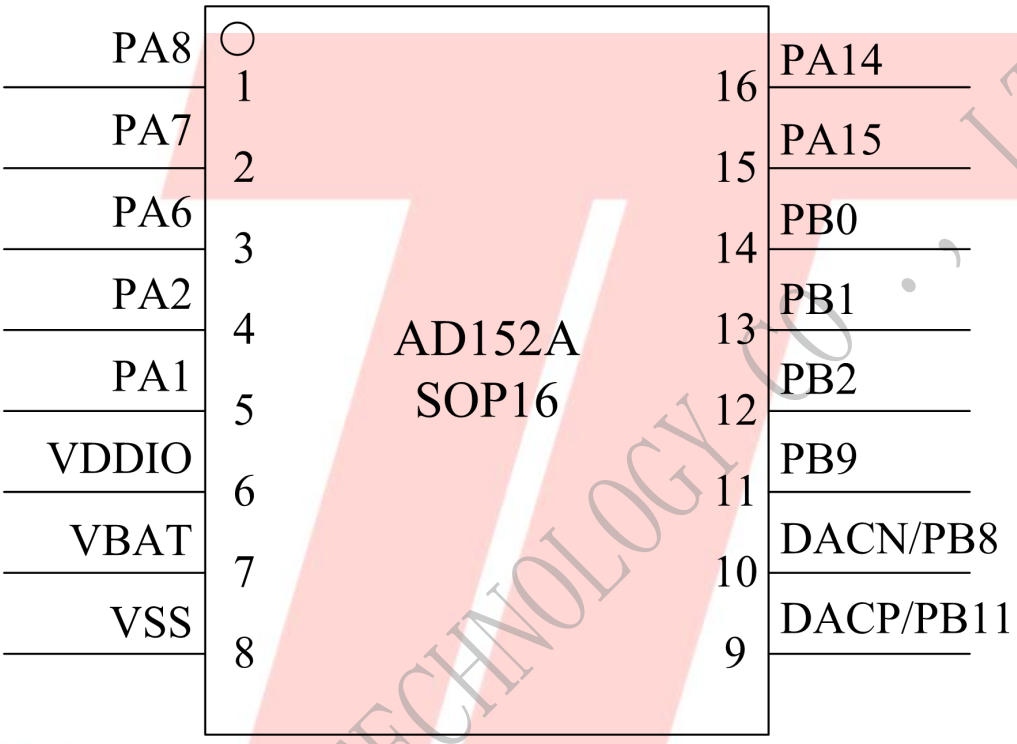

Figure 2-1 AD152A_SOP16 Package Diagram

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2.2 Pin Description

Table 2-1 AD152A_SOP16 Pin Description

PIN NO.	Name	Type	Drive (mA)	Function	Description
1	PA8	I/O	8/64	GPIO	SPI1DIC:SPI1 Data In(C); SD0DATD:SD0 Data(D);
2	PA7	I/O	8/64	GPIO	ADC7:ADC Input Channel 7; SPI1DOC:SPI1 Data Out(C); SD0CMDD:SD0 Command(D); UART0RXA:Uart0 Data In(A); I2C_SDA(C); PWM1:PWM Channel1 Output;
3	PA6	I/O	8/64	GPIO	ADC6:ADC Input Channel 6; SPI1CLKC:SPI1 Clock(C); SD0CLKD:SD0 Clock(D); UART0TXA:Uart0 Data Out(A); I2C_SCL(C); TMR2:Timer2 Clock In; PWM0:PWM Channel0 Output;
4	PA2	I/O	8/64	GPIO	ADC2:ADC Input Channel 2; SPI0DOB(0):SPI0 Data0 Out(B); I2C_SDA(B); PWM2(A);
5	PA1	I/O	8/64	GPIO	ADC1:ADC Input Channel 1; SPI0CLKB:SPI0 Clock(B); UART0RXB:Uart0 Data In(B); I2C_SCL(B); CAP2:Timer2 Capture;
6	VDDIO	P	/		Digital Power; (Internal linear regulator output)
7	VBAT	P	/		Battery Power Supply;
8	VSS	G	/		Ground;
9	DACP	O	/		Class-D APA Positive Output;
	PB11	I/O	8	GPIO (High Voltage Resistance)	OSCIB:Crystal Oscillator Input(B);
10	DACN	O	/		Class-D APA Negative Output;
	PB8	I/O	8	GPIO (High Voltage Resistance)	SPI1CLKD:SPI1 Clock(D); I2C_SCL(D); OSCIA:Crystal Oscillator Input(A);

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11	PB9	I/O	8	GPIO (High Voltage Resistance)	UART1TRXB:Uart1 Data In/Out(B); SPI1DOD:SPI0 Data Out(D); I2C_SDA(D); CAP1:Timer1 Capture;
12	PB2	I/O	8/64	GPIO	SPI1DIA:SPI1 Data In(A); SD0DATB:SD0 Data(B);
13	PB1	I/O	8/64	GPIO (pull down)	ADC11:ADC Input Channel 11; SPI1DOA:SPI1 Data Out(A); SD0CMDB:SD0 Command(B); I2C_SDA(A);
14	PB0	I/O	8/64	GPIO (pull down)	ADC10:ADC Input Channel 10; SPI1CLKA:SPI1 Clock(A); SD0CLKB:SD0 Clock(B); I2C_SCL(A);
15	PA15	I/O	8/64	GPIO	ADC9:ADC Input Channel 9; SPI1DOB:SPI1 Data Out(B); MCAP3:Motor Timer3 Capture;
16	PA14	I/O	8/64	GPIO	ADC8:ADC Input Channel 8; SPI1CLKB:SPI1 Clock(B); CAP0:Timer0 Capture; MCAP2:Motor Timer2 Capture;

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3 Electrical Characteristics

3.1 Absolute Maximum Ratings

Table 3-1

Symbol	Parameter	Min	Max	Unit
Tamb	Ambient Temperature	-40	+85	°C
Tstg	Storage temperature	-65	+150	°C
VBAT	Supply Voltage	-0.3	5.5	V
$V_{VDDIO33}$	3.3V IO Input Voltage	-0.3	3.6	V

Note : The chip can be damaged by any stress in excess of the absolute maximum ratings listed below.

3.2 PMU Characteristics

Table 3-2

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
VBAT	Voltage Input	2.0	3.7	5.5	V	_
V_{VDDIO}	Voltage output	2.0	3.0	3.4	V	VBAT = 3.7V, 100mA loading
I_{VDDIO}	Loading current	_	_	100	mA	VBAT=3.7V

3.3 IO Input/Output Electrical Logical Characteristics

Table 3-3

IO input characteristics						
Symbol	**Parameter**	**Min**	**Typ**	**Max**	**Unit**	**Test Conditions**
V_{IL}	Low-Level Input Voltage	-0.3	_	0.3* VDDIO	V	VDDIO = 3.3V
V_{IH}	High-Level Input Voltage	0.7* VDDIO	_	VDDIO+0.3	V	VDDIO = 3.3V
IO output characteristics						
V_{OL}	Low-Level Output Voltage	_	_	0.33	V	VDDIO = 3.3V
V_{OH}	High-Level Output Voltage	2.7	_	_	V	VDDIO = 3.3V

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3.4 Internal Resistor Characteristics

Table 3-4

Port	General Output	High Drive	Internal Pull-Up Resistor	Internal Pull-Down Resistor	Comment
PA0~PA12 PB0~PB7	8mA	64mA	10K	60K	1、PA0 default pull up 2、PB0 & PB1 default pull down 3、internal pull-up/pull-down resistance | accuracy ±20%
PB8,PB9, PB11	8mA	_	10K	60K	

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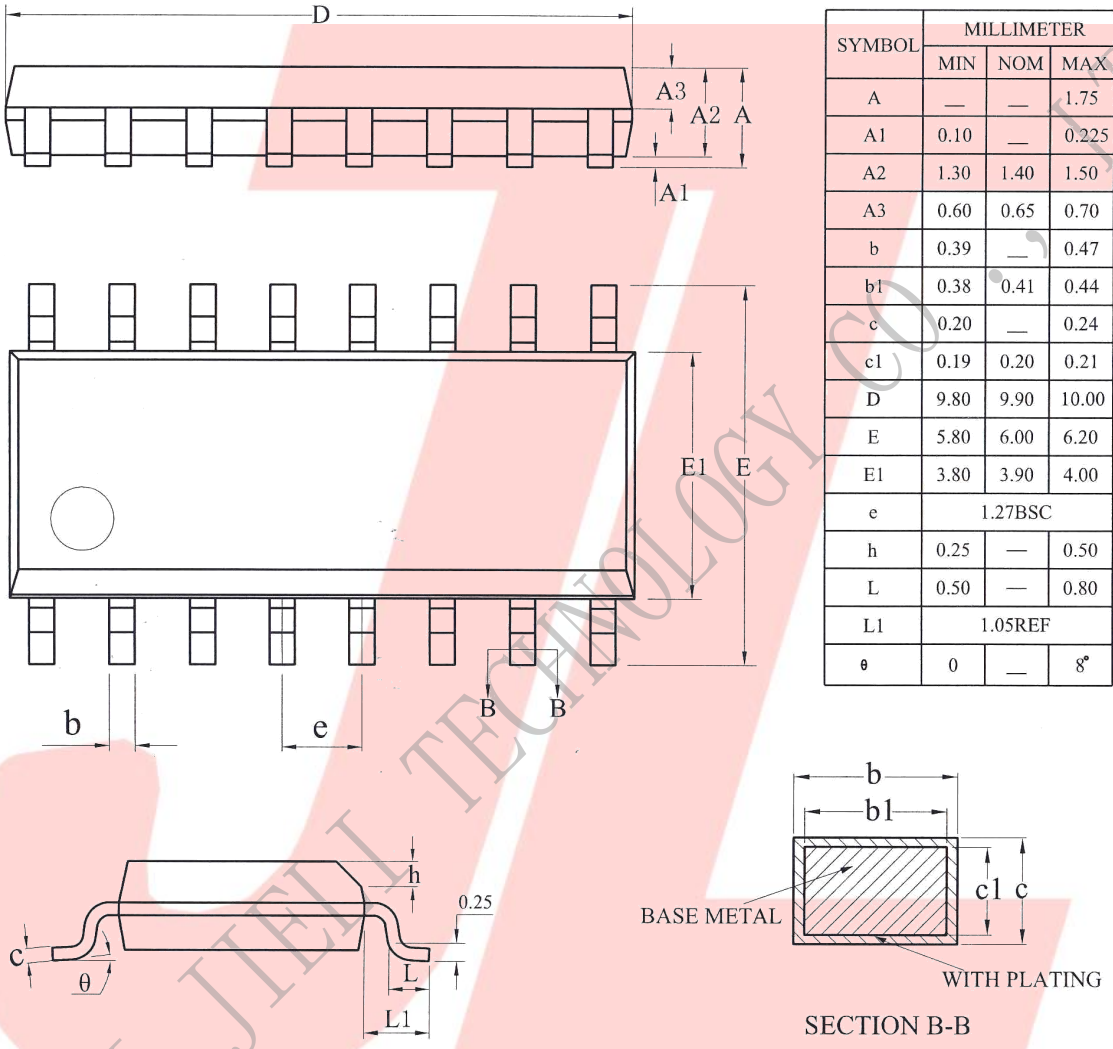
4 Package Information

4.1 SOP16



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	—	—	1.75
A1	0.10	—	0.225
A2	1.30	1.40	1.50
A3	0.60	0.65	0.70
b	0.39	—	0.47
b1	0.38	0.41	0.44
c	0.20	—	0.24
c1	0.19	0.20	0.21
D	9.80	9.90	10.00
E	5.80	6.00	6.20
E1	3.80	3.90	4.00
e	1.27BSC		
h	0.25	—	0.50
L	0.50	—	0.80
L1	1.05REF		
θ	0	—	8°

Figure 4-1. AD152A_SOP16 Package

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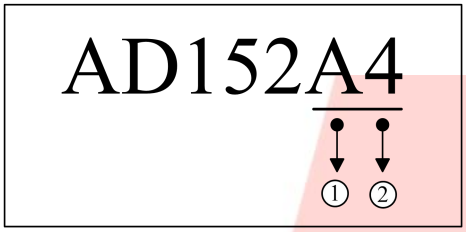
5 Package Type Specification



①Represents different packages

②Represents different memory sizes

0: No memory

2: 2Mbit Flash

4: 4Mbit Flash

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6 Storage Condition

6.1 Moisture Sensitivity Level

AD152A is qualified to moisture sensitivity level MSL3 in accordance with JEDEC J-STD-033.

6.2 Storage Alert

1. Calculated shelf life in sealed bag 12 months at＜40°C and 90% relative humidity (RH).
2. Peak package body temperature≤260°C.
3. After bag is opened,devices that will be subjected to reflow solder or other high temperature process must be mounted within 168 hours of factory conditions≤30°C/60%RH or stored per J-STD-033.
4. Devices require bake before mounting if humidity indicator card reads＞10% for level 2a-5a devices or＞60% for level 2 devices when read at 23±5°C,or 3a or 3b are not met.

Please refer to IPC/JEDEC J-STD-033 for baking procedure if necessary.

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7 Revision History

Date	Revision	Description
2021.03.18	V1.0	Initial Release.
2021.08.23	V1.1	Modify the pin definition.
2023.03.21	V1.2	Modify the Features.
2024.11.20	V1.3	Add Block Diagram Add Storage Condition Update Features modification.

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